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Highway to...

Determining Fatal Outcomes in Traffic Accidents Based on Police Reports

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BRACIS
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Introduction



- Brazil has the 4th largest road network worldwide
 - Over 1.7 million km of roads
- Crucial for economic integration and logistics
- But poor conditions & intense use
- Cause of many deaths annually

Motivation



pesquisafapesp

PUBLIC POLICY

Traffic deaths increase by 13.5% in the last decade

Roads are less safe in Brazil, largely due to motorcycle accidents, with road improvements and educational campaigns desperately needed

In 2020, an accident with five motorcycles and another 17 vehicles in São José dos Pinhais (Paraná State) left at least eight people dead and another 20 injured



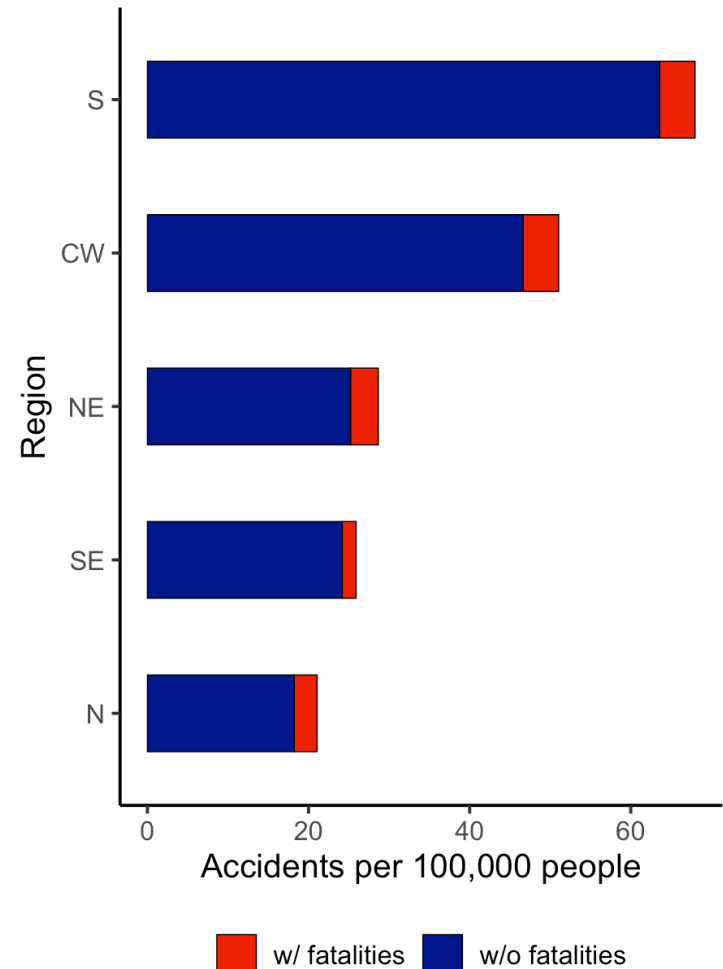
<https://revistapesquisa.fapesp.br/en/traffic-deaths-increase-by-13-5-in-the-last-decade/>

Motivation



In 2024 alone:

- Brazil registered:
 - 67,000+ accidents
 - R\$14bi+ of economic cost
 - 16 deaths per day
- In the Southern region:
 - 21,000+ accidents
 - R\$4.3bi+ of economic cost
 - 6.4% fatal



Research Objective



- Predict fatal vs. non-fatal accident outcomes
- Compare ML models: RF, kNN, MLP
- Employ an explainability method on best model
 - Extract actionable insights for policy making

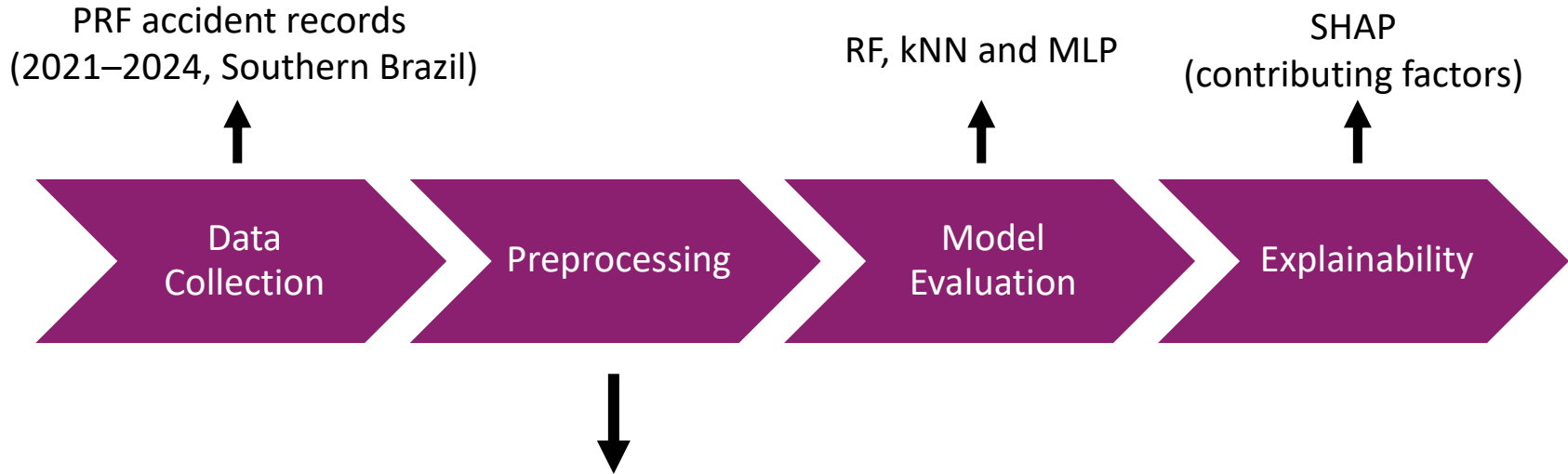
Dataset



- PRF open dataset (2021-24)
 - Southern region
- Accident reports including:
 - Demographics
 - Road conditions
 - Weather, ...
- Challenge
 - Mostly categorical features
 - Target class imbalance

Feature	Type
severity	target
type_accident	categorical
type_involved	categorical
age	numerical
gender	categorical
type_vehicle	categorical
year_vehicle	numerical
day_week	categorical
day_period	categorical
met_cond	categorical
type_road	categorical
out_road	categorical
id_road	categorical
km_road	numerical
direction_road	categorical

Methodology



- Cleaning inconsistent/missing values
- Categorical reduction + target encoding
- SMOTE to rebalance fatalities from 7.1% → ~50%

Model Selection



- Split 80/20 train-test + Monte Carlo CV
 - Repeated 80/20 train-validation five times
- Hyperparameter tuning (grid search)
- Metrics: ACC, TPR, PPV, F1-score, **AUC-ROC**

Model	Hyperparameter	Interval
kNN	k	[3, 5, 7, ..., 15]
RF	mtry; split rule; min. node size	[1, 2, ..., 12]; { <i>gini</i> , <i>extratrees</i> }; [1, 5, 10]
MLP	size; decay	[11, 19, 27, ..., 115]; $10^{[-4, -3, \dots, -1]}$

Model Performance



- Oversampling greatly reduced false negative rates

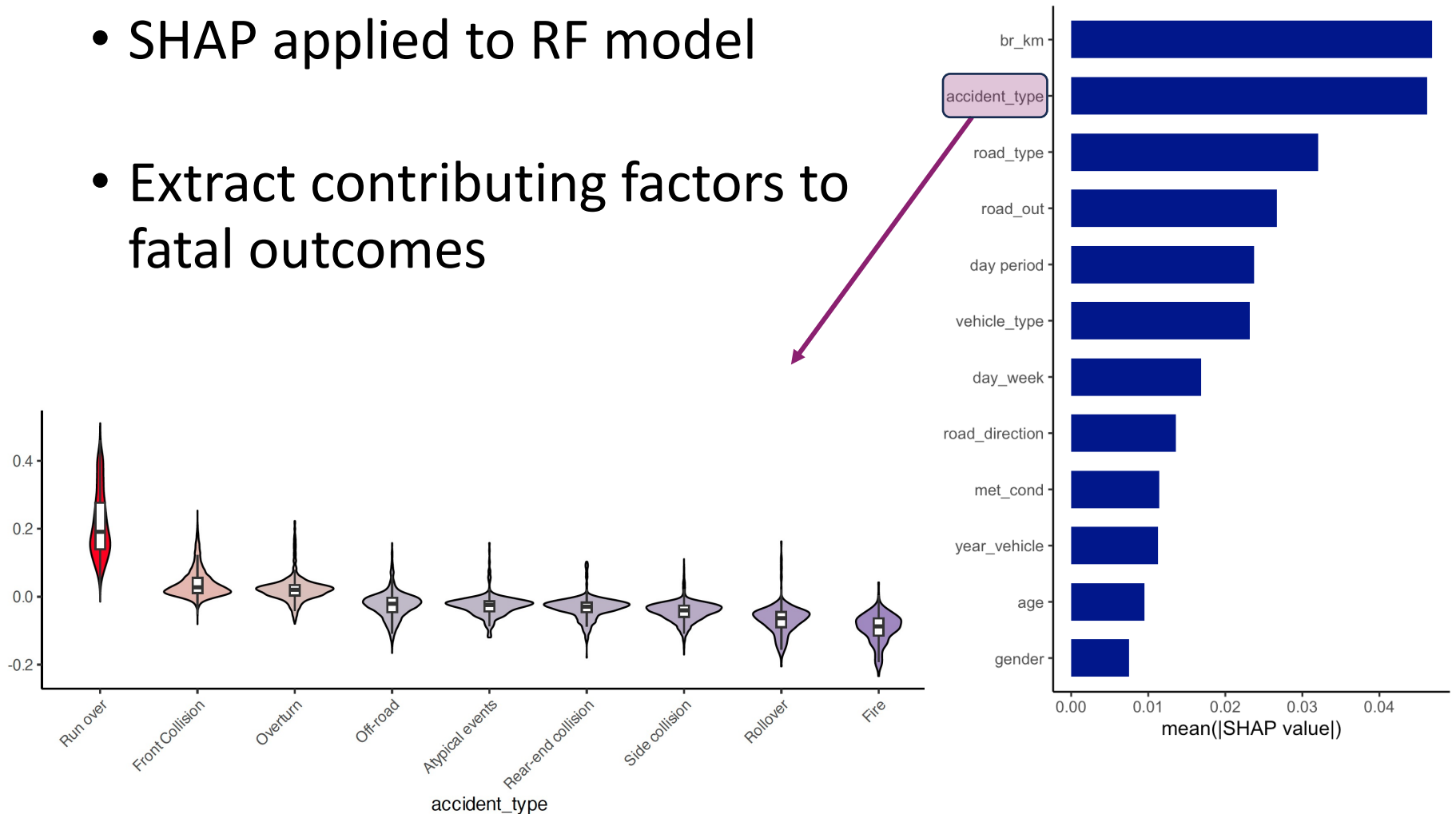
Model	Accuracy	Sensitivity	Specificity	AUC-ROC	F1-Score
w/o SMOTE					
kNN	0.94	0.19	0.99	0.92	0.30
RF	0.99	0.99	1.00	0.99	0.99
MLP	0.93	0.14	0.99	0.82	0.23

w/ SMOTE					
kNN	0.86	1.00	0.85	0.99	0.50
RF	0.99	0.99	0.99	0.99	0.99
MLP	0.77	0.71	0.77	0.82	0.29

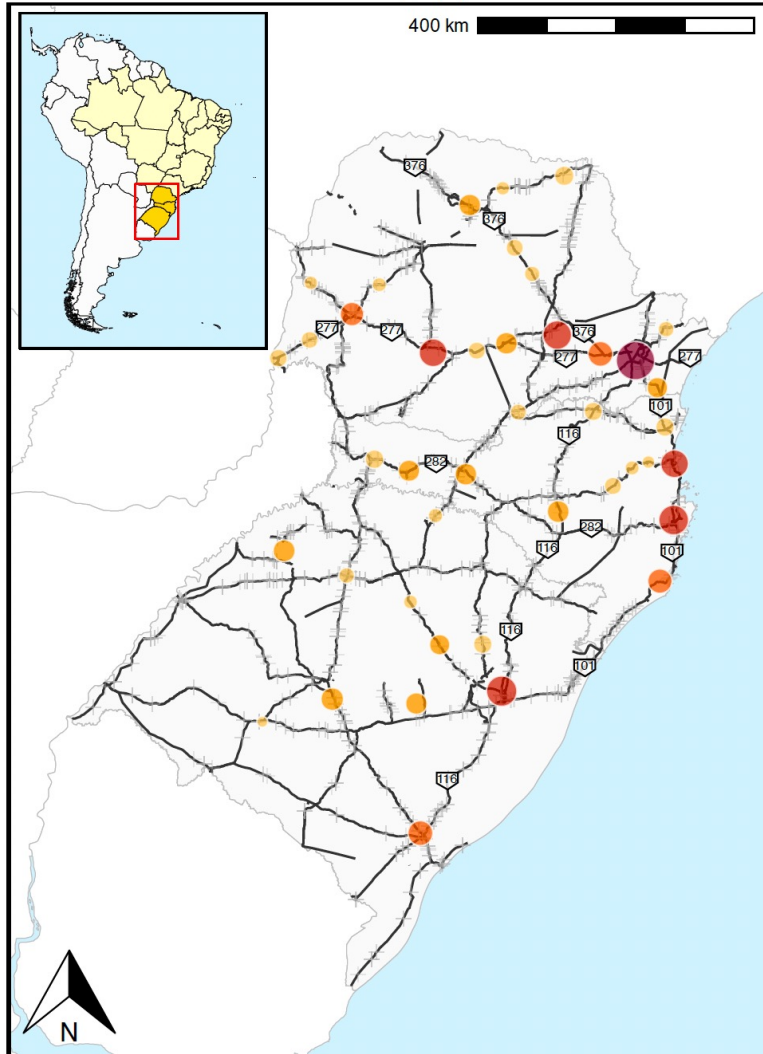
- RF was selected for explainability methods

Explainability

- SHAP applied to RF model
- Extract contributing factors to fatal outcomes



Spatial Insights



Hotspot highways:

- BR-277, BR-116, BR-376, BR-282, BR-101
- Structural problems:
 - single lanes
 - poor maintenance
 - urban crossings
 - cargo traffic

Final Remarks



- SMOTE proved efficient in reducing false negatives
- RF achieved high predictive performance
- SHAP analysis showed the models' capability to extract actionable knowledge
- Future work: expand analysis to all Brazilian regions
 - Study regional interplays and distinct contributing factors

Thank you for the attention!

Questions?

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Acknowledgements

